

Lemma: 对 $\forall x, y \in \mathbb{R}$, 有: $\max\{x, y\} = \frac{x+y+|x-y|}{2}$

Proof: 当 $x \geq y$ 时, 左边 $= \max\{x, y\} = x$

$$\text{右边} = \frac{x+y+x-y}{2} = \frac{2x}{2} = x. \quad \therefore \text{左边} = \text{右边}.$$

当 ~~x~~ $x < y$ 时, 左边 $= \max\{x, y\} = y$

$$\text{右边} = \frac{x+y+y-x}{2} = y \quad \therefore \text{左边} = \text{右边}. \quad \square$$

~~Lemma: 对 $\forall k \in \mathbb{R}, \forall x, y \in \mathbb{R}$, 有: $\max\{kx, ky\} = k \cdot \max\{x, y\}$~~

~~Proof: max~~

Lemma: 对 $\forall x, y \in \mathbb{R}$, 有: $\min\{x, y\} = \frac{x+y-|x-y|}{2}$

Proof: 当 $x \geq y$ 时, 左边 $= \min\{x, y\} = y$

$$\text{右边} = \frac{x+y-(x-y)}{2} = \frac{x+y-x+y}{2} = y \quad \therefore \text{左边} = \text{右边}.$$

当 $x < y$ 时, 左边 $= \min\{x, y\} = x$

$$\text{右边} = \frac{x+y+(x-y)}{2} = x \quad \therefore \text{左边} = \text{右边}. \quad \square$$

Lemma: 对 $\forall k, x, y \in \mathbb{R}$, 有:

$$\max\{kx, ky\} = \begin{cases} k \max\{x, y\} & , \text{当 } k \geq 0 \text{ 时} \\ k \min\{x, y\} & , \text{当 } k < 0 \text{ 时} \end{cases}$$

$$\begin{aligned} \text{Proof: 左边} &= \max\{kx, ky\} = \frac{kx+ky+|kx-ky|}{2} = \frac{kx+ky+|k(x-y)|}{2} \\ &= \frac{kx+ky+|k||x-y|}{2} \end{aligned}$$

当 $k \geq 0$ 时, 左边 $= \frac{kx + ky + k|x-y|}{2} = k \cdot \frac{x+y+|x-y|}{2} = k \cdot \max\{x, y\}$.

当 $k < 0$ 时, 左边 $= \frac{kx + ky - k|x-y|}{2} = k \cdot \frac{x+y-|x-y|}{2} = k \cdot \min\{x, y\}$ □

Lemma: 对 $\forall x, y \in \mathbb{R}$, 有: $\max\{|x+y|, |x-y|\} = |x| + |y|$

Proof: 若 $x=0$, 则左边 $= \max\{|y|, |-y|\} = \max\{|y|, |y|\} = |y|$
 $= 0 + |y| = |x| + |y| =$ 右边.

若 $y=0$, 则左边 $= \max\{|x|, |x|\} = |x| = |x| + 0 = |x| + |y| =$ 右边

$\therefore x=0$ 或 $y=0$ 时, 得证. 下设 $x \neq 0$ 且 $y \neq 0$.

① $x > 0$ 且 $y > 0$. 此时 $|x| = x$, $|y| = y$, $|x+y| = x+y$.

$\therefore |x-y| = |x+(-y)| \leq |x| + |-y| = |x| + |y| = x+y$

$\therefore$ 左边 $= \max\{|x+y|, |x-y|\} = \max\{x+y, |x-y|\} = x+y = |x| + |y|$
 $=$ 右边.

② $x > 0$ 且 $y < 0$. 此时 $|x| = x$, $|y| = -y$, $|x-y| = x-y$

$\therefore |x+y| \leq |x| + |y| = x + (-y) = x-y$.

$\therefore$ 左边 $= \max\{|x+y|, |x-y|\} = \max\{|x+y|, x-y\} = x-y = x + (-y)$
 $= |x| + |y| =$ 右边.

③ $x < 0$ 且 $y > 0$. 此时 $|x| = -x$, $|y| = y$, $|x-y| = y-x$.

$\therefore |x+y| \leq |x| + |y| = (-x) + y = y-x$

$$\therefore \text{左边} = \max\{|x+y|, |x-y|\} = \max\{|x+y|, y-x\} = y-x \\ = (-x) + y = |x| + |y| = \text{右边}.$$

④. $x < 0$ 且 $y < 0$. ~~同理可证~~

$$\text{此时 } |x| = -x, |y| = -y, |x+y| = -(x+y) = (-x) + (-y)$$

$$\therefore |x-y| = |x+(-y)| \leq |x| + |-y| = |x| + |y| = (-x) + (-y)$$

$$\therefore \text{左边} = \max\{|x+y|, |x-y|\} = \max\{(-x) + (-y), |x-y|\} \\ = (-x) + (-y) = |x| + |y| = \text{右边}. \quad \square$$

Lemma: 对 $\forall t, a, b \in \mathbb{R}$, 有:

$$\max\{|t-a|, |t-b|\} = \left|t - \frac{a+b}{2}\right| + \frac{|a-b|}{2}$$

Proof: 令 $\alpha = \frac{a+b}{2}$, $\beta = \frac{a-b}{2}$, 则有: $a = \alpha + \beta$, $b = \alpha - \beta$

$$\therefore \text{左边} = \max\{|t-a|, |t-b|\} = \max\{|t-\alpha-\beta|, |t-\alpha+\beta|\}$$

$$= \max\{|(t-\alpha)+\beta|, |(t-\alpha)-\beta|\} = |t-\alpha| + |\beta|$$

$$= \left|t - \frac{a+b}{2}\right| + \left|\frac{a-b}{2}\right| = \left|t - \frac{a+b}{2}\right| + \frac{|a-b|}{2} = \text{右边}. \quad \square$$

Lemma: 对 $\forall x, y \in \mathbb{R}$, 有: $|x+y| \leq |x| + |y|$

Lemma: 对 $\forall x, y, z \in \mathbb{R}$, 若 $x \leq z$ 且 $y \leq z$, 则有:
 $\max\{x, y\} \leq z$.

这两个引理的证明过于显然, 此处略去.

Lemma: 对 $\forall P(x_1, y_1) \in \mathbb{R}^2, Q(x_2, y_2) \in \mathbb{R}^2$, 有:

$d(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\}$ 是 $\mathbb{R}^2$ 上的一个度量.

Proof: 对 $\forall P, Q \in \mathbb{R}^2$, 设 $P(x_1, y_1), Q(x_2, y_2)$.

(其中 $x_1, y_1, x_2, y_2 \in \mathbb{R}$).

$$d(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\} = |x_1 - x_2| \text{ 或 } |y_1 - y_2| \geq 0.$$

$$d(P, Q) = 0 \Leftrightarrow \max\{|x_1 - x_2|, |y_1 - y_2|\} = 0$$

$$\Leftrightarrow |x_1 - x_2| = 0 \text{ 且 } |y_1 - y_2| = 0 \Leftrightarrow x_1 = x_2 \text{ 且 } y_1 = y_2 \Leftrightarrow P = Q.$$

$$\begin{aligned} d(P, Q) &= \max\{|x_1 - x_2|, |y_1 - y_2|\} = \max\{|x_2 - x_1|, |y_2 - y_1|\} \\ &= d((x_2, y_2), (x_1, y_1)) = d(Q, P). \end{aligned}$$

对 $\forall P, Q, R \in \mathbb{R}^2$, 设 $P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3)$.

$$\therefore d(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\} \geq |x_1 - x_2|$$

$$d(Q, R) = \max\{|x_2 - x_3|, |y_2 - y_3|\} \geq |x_2 - x_3|$$

$$\therefore |x_1 - x_3| = |x_1 - x_2 + x_2 - x_3| \leq |x_1 - x_2| + |x_2 - x_3| \\ \leq d(P, Q) + d(Q, R)$$

$$\therefore d(P, Q) = \max\{|x_1 - x_2|, |y_1 - y_2|\} \geq |y_1 - y_2|$$

$$d(Q, R) = \max\{|x_2 - x_3|, |y_2 - y_3|\} \geq |y_2 - y_3|$$

$$\therefore |y_1 - y_3| = |y_1 - y_2 + y_2 - y_3| \leq |y_1 - y_2| + |y_2 - y_3| \\ \leq d(P, Q) + d(Q, R)$$

$$\therefore d(P, Q) + d(Q, R) \geq |x_1 - x_3|, \quad d(P, Q) + d(Q, R) \geq |y_1 - y_3|$$

$$\therefore d(P, Q) + d(Q, R) \geq \max\{|x_1 - x_3|, |y_1 - y_3|\} = d(P, R)$$

$$\therefore d(P, R) \leq d(P, Q) + d(Q, R)$$

$\therefore d$ 是 $\mathbb{R}^2$ 上的一个度量. $\square$

~~Lemma: $C(a, b) \in \mathbb{R}$~~

Lemma: $A(\alpha, \beta) \in \mathbb{R}^2$, $r \in \mathbb{R}_{\geq 0}$, 对 $\forall (x, y) \in \mathbb{R}^2$, 有:

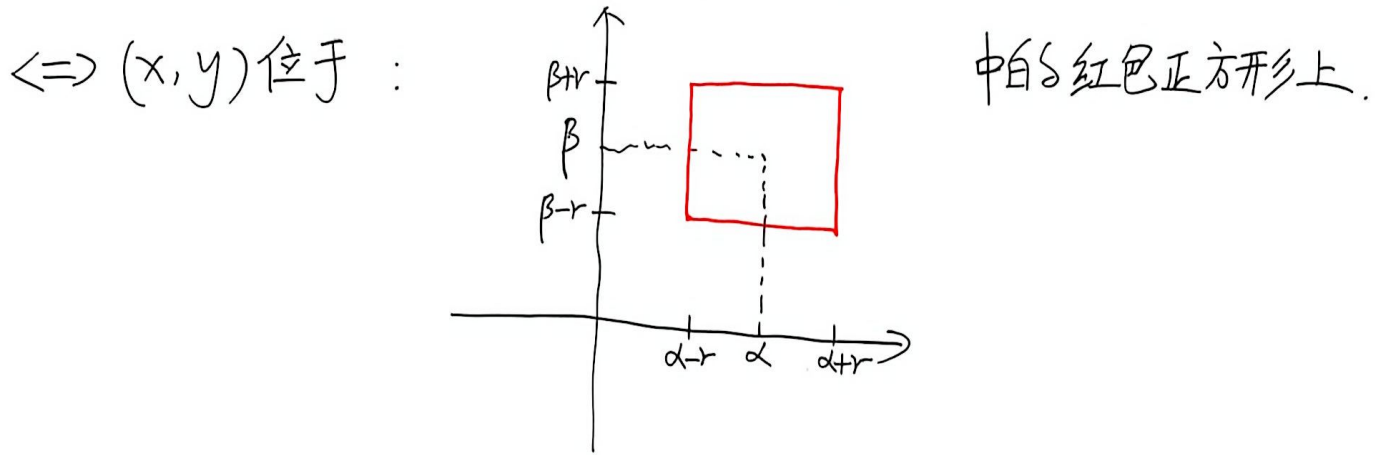
$$\textcircled{1} d((x, y), A) \leq r \Leftrightarrow \alpha - r \leq x \leq \alpha + r \text{ 且 } \beta - r \leq y \leq \beta + r$$

$$\textcircled{2} \text{ ~~} d((x, y), A) = r \Leftrightarrow (x = \alpha - r \text{ 或 } x = \alpha + r) \text{ 且 } (y = \beta - r \text{ 或 } y = \beta + r) \text{ }~~$$

$$\text{Proof: } \textcircled{1}: d((x, y), A) \leq r \Leftrightarrow \max\{|x - \alpha|, |y - \beta|\} \leq r$$

$$\Leftrightarrow |x - \alpha| \leq r \text{ 且 } |y - \beta| \leq r \Leftrightarrow \alpha - r \leq x \leq \alpha + r \quad \textcircled{1} \text{ 得证.} \\ \text{且 } \beta - r \leq y \leq \beta + r$$

$$\begin{aligned} \textcircled{2}: d((x, y), A) = r &\Leftrightarrow \max\{|x - \alpha|, |y - \beta|\} = r \\ &\Leftrightarrow (|x - \alpha| = r \text{ 且 } |y - \beta| \leq r) \text{ 或 } (|x - \alpha| \leq r \text{ 且 } |y - \beta| = r) \\ &\Leftrightarrow (x = \alpha \pm r \text{ 且 } \beta - r \leq y \leq \beta + r) \text{ 或 } (\alpha - r \leq x \leq \alpha + r \text{ 且 } y = \beta \pm r) \end{aligned}$$



□.

Lemma: ~~Let~~ $P(x_1, y_1) \in \mathbb{R}^2$, $Q(x_2, y_2) \in \mathbb{R}^2$, ~~Let~~

对 $\forall A(x, y) \in \mathbb{R}^2$ 满足 $d(A, P) = d(A, Q)$,

$d(A, P) = d(A, Q)$ 的最小值为 $\frac{1}{2}d(P, Q)$

Proof: 对 $\forall A(x, y) \in \mathbb{R}^2$ 满足 $d(A, P) = d(A, Q)$, 有:

$$\therefore d(P, Q) \leq d(P, A) + d(A, Q) = d(A, P) + d(A, Q)$$

$$\therefore d(P, Q) \leq 2d(A, P). \quad \therefore d(A, P) \geq \frac{1}{2}d(P, Q).$$

$$\therefore d(A, P) = d(A, Q) \geq \frac{1}{2}d(P, Q).$$

取 $A = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. 此时有:

$$\begin{aligned} d(A, P) &= d\left(\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), (x_1, y_1)\right) = \max\left\{\left|\frac{x_1 + x_2}{2} - x_1\right|, \left|\frac{y_1 + y_2}{2} - y_1\right|\right\} \\ &= \max\left\{\left|\frac{x_1 - x_2}{2}\right|, \left|\frac{y_1 - y_2}{2}\right|\right\} = \max\left\{\frac{1}{2}|x_1 - x_2|, \frac{1}{2}|y_1 - y_2|\right\} \end{aligned}$$

$$= \frac{1}{2} \max \{ |x_1 - x_2|, |y_1 - y_2| \} = \frac{1}{2} d(P, Q).$$

$$d(A, Q) = d\left(\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right), (x_2, y_2)\right) = \max \left\{ \left| \frac{x_1+x_2}{2} - x_2 \right|, \left| \frac{y_1+y_2}{2} - y_2 \right| \right\}$$

$$= \max \left\{ \left| \frac{x_1 - x_2}{2} \right|, \left| \frac{y_1 - y_2}{2} \right| \right\} = \max \left\{ \frac{1}{2} |x_1 - x_2|, \frac{1}{2} |y_1 - y_2| \right\}$$

$$= \frac{1}{2} \max \{ |x_1 - x_2|, |y_1 - y_2| \} = \frac{1}{2} d(P, Q).$$

$$\therefore d(A, P) = d(A, Q) = \frac{1}{2} d(P, Q).$$

$\therefore$ 当点 A 取为 P, Q 中点时, $d(A, P) = d(A, Q) = \frac{1}{2} d(P, Q)$ 成立.

□